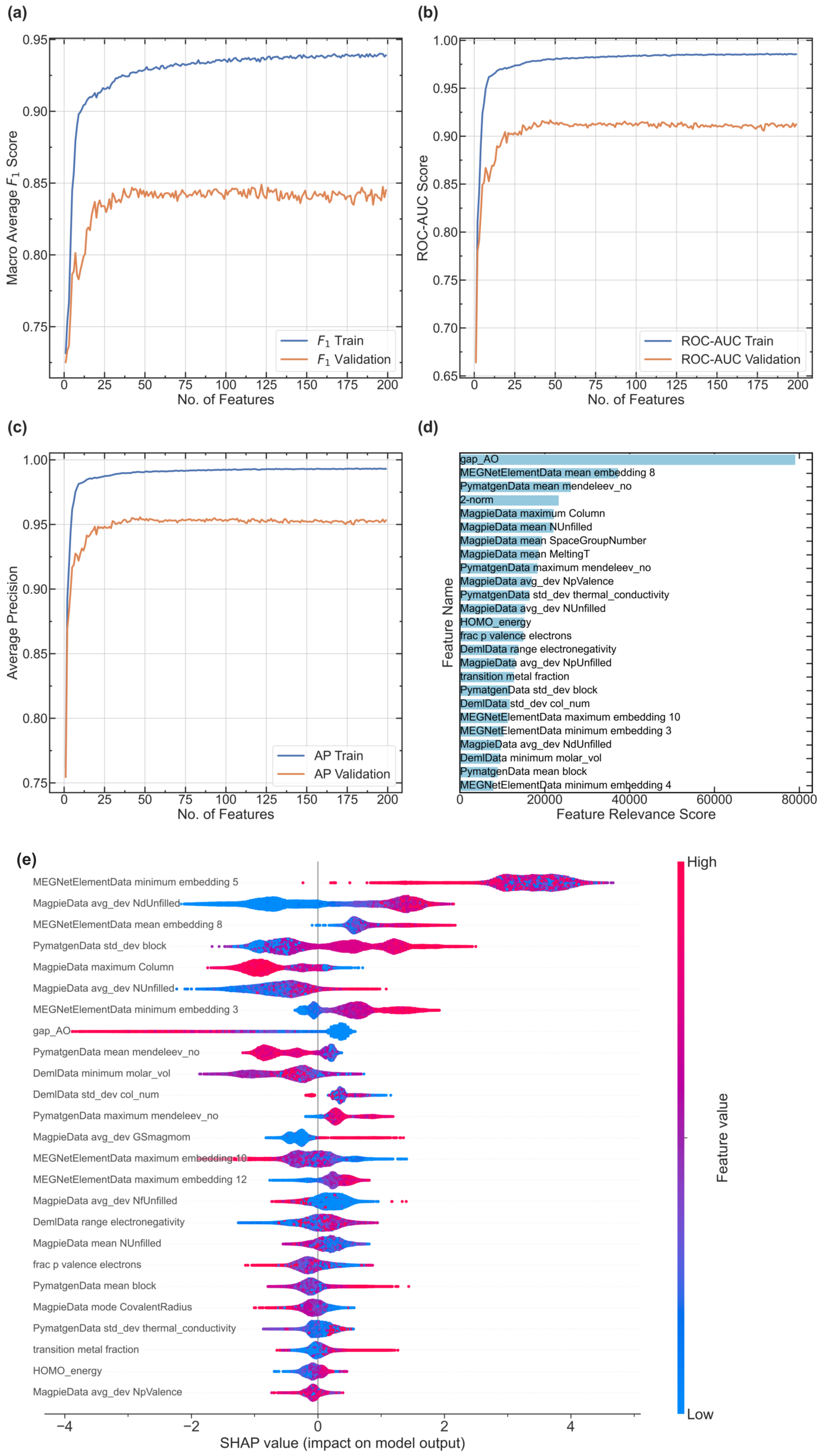
(a)
Macro Average F_1 Score
No. of Features
F_1 Train
F_1 Validation
(b)
ROC-AUC Score
No. of Features
ROC-AUC Train
ROC-AUC Validation
(c)
Average Precision
No. of Features
AP Train
AP Validation
(d)
Feature Name
Feature Relevance Score
gap_AO
MEGNetElementData mean embedding 8
PymatgenData mean mendeleev_no
2-norm
MagpieData maximum Column
MagpieData mean NUnfilled
MagpieData mean SpaceGroupNumber
MagpieData mean MeltingT
PymatgenData maximum mendeleev_no
MagpieData avg_dev NpValence
PymatgenData std_dev thermal_conductivity
MagpieData avg_dev NUnfilled
HOMO_energy
frac p valence electrons
DemlData range electronegativity
MagpieData avg_dev NpUnfilled
transition metal fraction
PymatgenData std_dev block
DemlData std_dev col_num
MEGNetElementData maximum embedding 10
MEGNetElementData minimum embedding 3
MagpieData avg_dev NdUnfilled
DemlData minimum molar_vol
PymatgenData mean block
MEGNetElementData minimum embedding 4
(e)
MEGNetElementData minimum embedding 5
MagpieData avg_dev NdUnfilled
MEGNetElementData mean embedding 8
PymatgenData std_dev block
MagpieData maximum Column
MagpieData avg_dev NUnfilled
MEGNetElementData minimum embedding 3
gap_AO
PymatgenData mean mendeleev_no
DemlData minimum molar_vol
DemlData std_dev col_num
PymatgenData maximum mendeleev_no
MagpieData avg_dev GSmagmom
MEGNetElementData maximum embedding 10
MEGNetElementData maximum embedding 12
MagpieData avg_dev NfUnfilled
DemlData range electronegativity
MagpieData mean NUnfilled
frac p valence electrons
PymatgenData mean block
MagpieData mode CovalentRadius
PymatgenData std_dev thermal_conductivity
transition metal fraction
HOMO_energy
MagpieData avg_dev NpValence
SHAP value (impact on model output)
High
Feature value
Low